

UTKARSH DUDEJA

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SUMMARY

Applied Research Engineer with 2.7 years of experience turning machine learning research into production-grade systems. I work across the full lifecycle—from prototyping and experimentation to optimizing and deploying models at scale and possess deep hands-on experience in distributed data processing and GPU-accelerated computing. I am an IEEE-published researcher with strong mathematical foundations. I'm drawn to problems where rigorous research meets real-world deployment. Currently seeking applied research and AI/ML engineering roles.

SKILLS

ML & Deep Learning | PyTorch (DDP), TensorFlow, Keras, Scikit-learn, NumPy, LSTM, CNN, Transformers, Anomaly Detection, Time-Series Modeling

Advanced ML | Continual Learning, Model Compression (Quantization, Pruning)

NLP & GenAI | LangChain, LLMs, Hugging Face, RAG, Vector Databases, Sequence Tagging, Text Classification, LoRA, QLoRA

MLOps & Infrastructure | Docker, Kubernetes, Ray, Ollama, Redis, OpenSearch, MLflow, CI/CD Pipelines, Cassandra, Linux, Github

Data Engineering | PostgreSQL, Pyspark, Stream & Batch Processing Pipelines, FastAPI, GraphQL, Event-driven Architecture, Distributed Systems

Languages | Python, SQL, Rust, Golang

Networking & Protocols | TCP/IP, Modbus, MQTT, OPC-UA, Wireshark, Suricata, Zeek

PROFESSIONAL EXPERIENCE

WhizHack Technologies Pvt. Ltd.

Jan 2024 – Present

Applied Research Engineer

Apr 2026 – Present

- **Researched and developed a novel multi-layer behavioral anomaly detection system** that enables the **detection of zero-day threats** like silent device substitution. Optimized the system, **achieving a 5.4× throughput gain** and **90% reduction in median latency** under real production workloads. Solution was **endorsed by the CTO** and **became the foundation for a research paper and patent filing**.
- Built a causal-attribution and counterfactual analysis engine that **root-causes incidents** and generates **“what-if”** reasoning, correlating related anomalies across time and device pairs to surface **multi-stage activities**.
- Built an out-of-distribution (OOD) **evaluation framework** that exposed generalization gaps hidden by tied test-split metrics, **driving model selection and cutting false positives ~5× on novel attacks** — including obfuscation-based SQL-injection evasion. Tracked experiments and model versions in **MLflow**.

Associate Software Development Engineer

Apr 2024 – Mar 2026

- Engineered a fully automated CVE intelligence pipeline processing **26 years** of vulnerability data with **100% ingestion reliability** to extract structured threat insights from unstructured vulnerability descriptions.
- **Re-architected** the asset discovery system with a modular, event-driven design, cutting execution time from **~3 hours to ~5 minutes (98% faster)** and enabling **zero-downtime deployments** across **10+** latency-sensitive production environments.
- **Architected a distributed, high-performance deep packet analysis system** featuring adaptive packet capture, producer-consumer architecture with dynamic worker allocation, and parallel protocol parsing across **20+ industrial protocols**, **delivering ~60% faster processing in production**.
- **Optimized and maintained a real-time threat intelligence system** with automated multi-source ETL extraction (MITRE, AbuseIPDB, TOR, Feodo, ThreatFox), GraphQL API layer, and concurrent bulk lookup — **serving enriched threat data via FastAPI with Redis caching**.

Software Development Intern

Jan 2024 – Mar 2024

- **Contributed to the core R&D team** focused on developing security systems; gained hands-on experience with OpenSearch, Fluentd, and Kibana for large-scale data analysis and visualization.

PROJECTS

Distributed ML Recommendation System | Kafka, Spark, PyTorch DDP, Ray Serve, Redis

- Built an end-to-end distributed ML system with **real-time exactly-once streaming**, **multi-node training**, and **fault-tolerant scalable serving**; sustained **382 req/s at 190ms p50** with a **97.5% cache hit rate**, and validated recovery from broker, consumer, Spark, and replica failures via injected-fault tests with **zero data loss**.

ATLAS — Autonomous Multi-Source Threat Correlation & Response Agent | Raspberry Pi 5, Ollama, Qwen 3.5, Langchain, CTI

- Built an autonomous SOC triage agent using LangGraph and a locally hosted LLM that correlates Wazuh and Suricata telemetry, enriches indicators with threat intelligence, and drives automated response decisions behind prompt-injection guardrails.

PUBLICATIONS

Decoding Emotions: Emotion Classification from EEG Brain Signals using AI

— IEEE Xplore

Accurate Object Detection for Intelligent Self-Driving Car System for Mountainous Terrains

— Forthcoming

EDUCATION & CERTIFICATIONS

B.Tech in CSE (Hons. Specialization in AI & ML) — Amity University, Uttar Pradesh

2020 – 2024

Duolingo English Test - Advanced: CEFR C1 (135/160) [Link](#)